

Research Article

Charting a New Course: An Inventory of Data Visualization Scholarship in TPC From 2013 to 2023

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Abstract—Background: This article presents a systematic review of data visualization scholarship in technical and professional communication (TPC) from 2013 to 2023. For interdisciplinary fields, such as TPC, where scholars draw on a wide range of epistemologies and methodologies, identifying patterns and trends prevents scholars from unknowingly talking past each other. **Literature review:** Our study was grounded in categorization scholarship in technical communication and contemporary issues the field faces. **Research questions:** 1. What kinds of articles are being published on data visualizations in TPC? 2. What case studies or examples do TPC scholars use to study data visualizations and from which disciplines or domains do they draw? 3. How can we categorize the types of data visualization written about in TPC journals? 4. What can these findings tell us about the current state of data visualizations in TPC journals and where further attention might be needed? **Methodology:** We conducted a content analysis of 94 articles across seven leading TPC journals. We developed a code book to identify common subject themes across scholarship looking at various types of data visualization types and topics. **Results:** Our results explore the field's growing emphasis on socially engaged themes, such as health, risk, and environmental justice. **Conclusion:** We call TPC to turn attention to accessibility and pedagogy.

Index Terms—Accessibility, COVID-19, data visualization, mapping, pedagogy, social justice.

Categorization scholarship provides a critical framework for understanding the organization and cohesion of scholarly activity within disciplinary fields. For interdisciplinary fields, such as technical and professional communication (TPC), where scholars draw on a wide range of epistemologies and methodologies, identifying patterns and trends prevents scholars from unknowingly talking past each other. Melonçon [1, p. 24] describes this fragmentation as a “methodological impasse,” where different disciplines struggle to reconcile varying approaches of collecting evidence and research design. For example, Melonçon and St.Amant mapped and categorized empirical research articles published in TPC journals from 2011 to 2015 to create a “picture of the field’s identity” based on its research methodologies [2, p. 133].

Efforts to systematically catalog scholarship also mitigate the incommensurability that Melonçon and St.Amant [2] identify as a threat to the

legitimacy of TPC. For example, a detailed categorization of scholarship within the rhetoric of health and medicine highlights both the opportunities and limitations that can emerge from an interdisciplinary field [3], as this subfield continues to grow and respond to the current moment. Some scholars have applied this methodological approach to data visualization scholarship when reviewing accessible design practices [4]. In data visualization scholarship, fragmentation may contribute to siloed innovation, underdeveloped pedagogical or theoretical frameworks, and limited citation practices. By following the model these scholars have laid out, we provide a categorization of data visualization scholarship to effectively identify existing trends and which areas in data visualization need more scholarly attention.

TPC researchers have provided valuable insights into large-scale disciplinary patterns [1], and others have taken a more granular approach to looking at the use of visuals in TPC journals [4]. The time is right for another look at the state of data visualizations in TPC. From 2013 to 2023, technical communication (TC) scholars interested in data visualizations have worked against the backdrop of scholarly changes, political uncertainty, and collective trauma. TPC has been profoundly shaped by the social justice turn [5], the COVID-19 pandemic, the Black Lives Matter

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Practitioner Takeaway

- Designers should prioritize accessibility considerations by incorporating inclusive design elements like high-contrast colors, alternative text, and screen reader compatibility when developing data visualizations.
 - Instructors should focus on developing pedagogy that teaches student-users to recognize how data visualizations can mislead so that they can be better equipped to sift through misinformation and rhetoricized data.
 - Practitioners should design data visualizations with an understanding of the kinds of perception, cognition, misinformation, and literacy issues audiences experience when engaging with data visualizations.
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movement, the proliferation of misinformation, and the advent of generative artificial intelligence. As such, there is a need for new scholarship to look at how technical communicators have specifically responded to the aforementioned shifts.

To categorize data visualization scholarship, we must first identify our terms. We define data visualizations as graphical representations of information—such as charts, graphs, maps, or infographics—designed to communicate data in a visual and synoptic, “at-a-glance” [6, p. 143] way. We also define data literacy as understanding that “Data visualization, then, does not just involve representing a given set of numbers; it involves selecting and rethinking the numbers on which the visualization is based” [7, p. 347]. Then, in the “Literature Review” section, we highlight researchers, who have illuminated issues of data visualization and literacy, both within TPC [8], [9], [10] and in other disciplines [11], [12].

We began this categorization scholarship by investigating where data visualization research has been published and which primary topics have been focused on. This disciplinary inventory-taking enables a deeper understanding of the field’s trajectory while addressing areas needing more scholarly attention. With the sunset of *Technical Communication*, we sought to take advantage of this kairotic moment to examine patterns in data visualization scholarship from 2013 to 2023. As data visualizations become even more popular, technical communicators must attend to issues of misinformation, literacy, pedagogy, and dimensions of accessibility to take part in conversations about health decision making [13], climate change [14], and demographics [15].

This article conducts a systematic review of data visualization literature from 2013 to 2023 to map the field’s scholarly activity. In our review of 94

articles, we identify which types of data visualizations are written about, which journals support this research, and the most frequent thematic content. While the field increasingly embraces social justice and human-centered themes—including risk, health, and environmental concerns—issues of accessibility and pedagogy require more attention. By critically reflecting on these patterns, we encourage scholars to prioritize research on accessibility’s visual, interactive, and cognitive dimensions.

LITERATURE REVIEW

Our literature review explores citation practices in the field and presents a review of contemporary challenges data visualization scholarship faces, including issues related to literacy, misinformation, accessibility, and rhetoricized data.

Interdisciplinary Data Visualization Scholarship Before 2013

Research on data visualization in TPC owes much to statisticians, journalists, and computer scientists. Before 2013, Tufte [11] published books examining and explaining the visual affordances of historically important charts and maps, such as Minard’s chart of Napoleon’s invasion of Russia (also discussed in more humanistic terms within TPC by Dragga and Voss [9]). TPC scholars have also drawn from the thoroughly researched historical scholarship conducted by the statisticians Friendly and Wainer [12] and physician Hans Rosling’s ability to convert massive datasets into meaningful data stories about global public health [16]. These interdisciplinary researchers paved the way for TPC articles published between 2013 and 2023.

Present Problems in Data Visualization TPC faces unprecedented challenges related to user literacy as the digital landscape becomes increasingly cluttered with misinformation and

deceptive design practices. Scholarship points out that data visualizations can be strategically designed to distort reality, shape public perception, and spread misinformation, since we saw with charts during the COVID-19 pandemic [17]. However, TPC scholars assert that visualizations greatly influence our perception of truth [18], but readers are rarely equipped with the literacy skills to critically engage with them. In their study on deceptive tactics in charts and graphs, Lauer and O'Brien [10, p. 334] found that "participants who cited previously taking a data visualization course were equally susceptible to deceptive tactics as those who had not." This is worrying in its own right but also indicates potential gaps in our pedagogical approach as technical communicators. When integrating eye-tracking approaches to learn more about how people visually process line graphs, Lauer and Sanchez found that participants did not spend much time looking at x - and y -axes in line graphs, especially when they had exaggerated titles, underscoring truncated scales and misleading titles as common ways readers are deceived. In addition, participants tended to be drawn in from the "visual interest created by the steeper slope," indicating that readers give more attention to line graph shape than the axes [19, p. 231]. Even small design choices can have large cognitive consequences for readers, creating greater challenges for technical communicators and readers alike.

TPC scholarship illuminates how data visualizations can emphasize certain data, arguments, or narratives through rhetoricized data [7]. Data can be represented in ways that are intentionally misleading, potentially obscuring the experiences of those that are most vulnerable [20]. The separate but related issues of misinformation and transparency around data's rhetorical nature [21], [22], [23] complicate our ability to evaluate sources critically, eroding public trust, and making it harder to collectively agree on what is true.

Technical communicators have a responsibility to ensure that their data visualizations are accessible, both visually and cognitively. Accessibility is defined as "the concept of whether a product or service can be used by everyone—however they encounter it" [24]. Visually, scholarship on data visualizations shows how they can pose specific challenges for disabled users, such as "color/contrast difficulties, size/scale issues, and inaccessibility to users with visual impairments" [4, p. 289]; hence, the Web Content Accessibility

Guidelines are a valuable resource for creating accessible data visualizations. Accessibility also has a cognitive dimension [10], [19], because neurodivergent users may need additional support to process visualized information, using a takeaway title or clear labeling can support these users' needs while reducing the potential for misinterpretation.

The articles we included in this study emphasized that challenges posed by misinformation and the rhetorical positioning of data are serious, not just for technical communicators but for democratic society more broadly. However, technical communicators are uniquely positioned to respond to these challenges by using data visualizations as tools for creating democratic consensus [25], [26]. In addition, educators in the field can cultivate critical data literacies by helping students question how data visualizations are constructed. Wolfe [7, p. 345] writes that textbooks on data visualization design overlook "the need to return to the data to reconsider what data are selected, how they are summarized, and whether they should be synthesized with other data for a more compelling argument." Rhetoricized data acknowledge that data always reflect a rhetor's point of view—purely neutral data, such as "objective" technical communication, do not exist. TPC scholars have risen to the occasion and used their expertise to promote critical engagement and transparent data communication.

Current Interdisciplinary Trends in Data Visualizations

These current trends expand beyond the disciplinary boundaries of TPC; journals, such as IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, push the boundaries of Big Data, digitized data, and visual analytics. Journalism professor Alberto Cairo's [18] work to analyze visual misinformation aligns with TPC's work on cognition and data literacy. Thorp's [27] exhortation to examine where data come from complements the field's work on rhetoricized data. Bremer and Wu bring data visualizations into everyday life [28], while McCandless [29] uses data visualizations to illuminate current issues in the news. TPC's work on data visualization connects to the digital humanities by thinking through feminist issues of equity and embodiment [30], [31] and digital accessibility through front-end engineering [32]. TPC owes much to this rich interdisciplinarity, while uniquely viewing data visualizations as an important part of audience-centered, rhetorical visual communication.

RESEARCH QUESTIONS

To guide our study, we developed research questions inspired by Melonçon and St.Amant [2] to map the scope and direction of data visualization scholarship in TPC.

RQ1. What kinds of articles are being published on data visualizations in TPC?

RQ2. What case studies or examples do TPC scholars use to study data visualizations and from which disciplines or domains do they draw?

RQ3. How can we categorize the types of data visualization written about in TPC journals?

RQ4. What can these findings tell us about the current state of data visualizations in TPC journals and where further attention might be needed?

METHODOLOGY

Our methodology was designed to systematically capture how data visualizations are discussed across a decade of TPC scholarship. To understand how scholars have organized and defined scholarly subfields, including data visualization, we conducted a content analysis of 94 peer-reviewed journal articles to identify publication patterns.

Data Collection Our data-collection process and broader methodological approach echo that of Melonçon and St.Amant's [2] method for tracking citations by collecting and reading relevant articles to look for overarching patterns and quantifying them through qualitative content analysis [33]. For this study, we looked at seven journals:

- *Journal of Business and Technical Communication* (JBTC)
- *Journal of Technical Writing and Communication*
- *Technical Communication Quarterly* (TCQ)
- *IEEE TRANSACTIONS ON PROFESSIONAL COMMUNICATION* (IEEE)
- *Communication Design Quarterly* (CDQ)
- *Business and Professional Communication Quarterly* (BPCQ)
- *Technical Communication*

For each journal, we reviewed articles published between January 2013 and December 2023, excluding book reviews or conference proceedings. We did, however, include the introductions of special issues for those that dealt closely with data visualization.

Our keywords included data visualizations and related terms, such as charts, graphs, maps, infographics, visual rhetoric, and visualization techniques. We examined all data visualization scholarship published from 2013 to 2023, in the seven TPC journals described above. When selecting articles, we relied on titles and keywords in the title and abstract and on the journal's website. In addition, articles that had been included based on their keywords were eliminated if they did not explicitly engage with data visualization. For example, an excellent article from the 2021 special issue of *JBTC* on COVID-19 was tagged as a data visualization article but did not engage with data visualizations in the text [34]. We defined data visualizations as charts, graphs, maps, or infographics that communicated data "at-a-glance" or synoptically [6, p. 143]. Most of our dataset explored more than one type of data visualization, and we coded each type that was central to an article's analysis. This expanded search yielded a total of 94 articles about data visualizations.

Coding In our first round of coding, each author separately open-coded the first 33 articles to inductively generate codes. At this point, we compared codes and created a codebook, which standardized the way that we coded, but also left room for us to add codes as needed. The categories we discuss in this article include the type of data visualization, the type of research, and engagement with data visualization. For the type of data visualization, we used Rebecca's [35] "Data Visualization Catalogue" to classify the types of data visualization we collected, including 12 articles about tables. We also built on Rebecca's definitions. For example, mapping was a significant type of data visualization in our dataset, but early on, we decided to differentiate between cartographic maps and process maps, as mapping a physical space and using a visual metaphor to overview a procedure have distinct design traits and rhetorical functions. Dashboards contained at least one data visualization that was updated in real time and usually interactive. In addition, some articles in the dataset did not focus on specific data visualizations, so we used the code "visualizations/general" to take note of them. Finally, we defined "infographics" to include diagrams and scientific illustrations with heavy amounts of data, as these are ways of visualizing data and processes and sometimes connect with isotype. We coded 24 types of charts, from activity trackers for menstrual cycles to waterfall charts.

We also coded for the genre of article: case study, editorial, essay, participant study, and tutorial. Finally, we coded for engagement with data visualizations. The first round of coding contained 25 possible codes for engagement with data visualizations; during the second round, we added three more.

During the second round of coding, we individually recoded the first 33 articles and the remaining 61 articles from the established codebook, as visual material often requires a holistic, language-based lens found in qualitative content analysis [33]. We compared our codes to check for consistency and consensus, discussing any disagreements and adding or subtracting codes across the four categories. In the “Results” section, we describe results from the three sections of our codebook: data visualization type, type of article, and the 15 (out of 28) most prevalent codes for engagement with data visualizations.

RESULTS

This section describes the results from coding 94 published data visualization articles from 2013 to 2023, based on their subject matter. Our results explore data visualization publications by article genre, year, journal, and code. COVID-19 had a significant impact on the scholarship published on data visualization, with 2021 marking the largest number of publications in the last decade. Finally, most articles in our dataset focused on themes related to the use of color, risk communication, user experience (UX), and health and disease.

Data Visualizations By Article Genre We coded six different article genres among all of these 94 articles: case studies, editorials, essays, literature reviews, participant studies, and tutorials. We found 47 case studies, which studied a particular issue or set of related issues without research participants, such as an analysis of “flatten the curve” visuals [25]. The three editorials opened the issue of a journal, particularly special editions, to focus on big-picture impacts of data visualizations [36], [37]. Five essays appeared that were not editorials, such as responses to research articles [38]. One literature review was published [39]. A total of 36 participant studies were included in the dataset, defined as researchers conducting research with participants outside the research team [10], [40]. Finally, we found two tutorials, both in IEEE TRANSACTIONS ON PROFESSIONAL COMMUNICATION.

Data Visualizations By Year The number of articles published on data visualization remained relatively consistent throughout the decade until the onset of the COVID-19 pandemic, as shown in Fig. 1. Overall, TPC journals have published around eight data visualization articles per year. These trends demonstrate a compelling and active subfield within TPC scholarship.

Special issues have been an important part of publishing activity within the field. In 2015, Kostelnick [36] edited a special issue in TCQ about data visualizations, marking a significant uptick in publications. In 2021, JBTC [37] published a special issue on COVID-19, which quickly invited many scholars to write about the ways that data visualizations were used in March and April 2020. This issue was initially available online in August 2020 and was later officially published in 2021. Of particular interest is the way that this special issue adopts “data visualization” as its own section. In 2022, COVID-19 still had an impact on the kinds of scholarly work published on data visualization, with only a slight decrease in interest in 2023. The narrative that this chart tells is that the field of data visualization has drawn more interest over the years, reaching a peak when health and disease visualizations became a popular site for critical analysis during the COVID-19 pandemic.

We noticed a peak in scholarship about Big Data in 2013–2016 (nine articles) compared to 2016–2023 (three articles). In addition, we noticed that articles pertaining to the importance of social justice, race, and culture emerged within TPC in the mid-2010s with the social justice turn. In the tradition of Walton, Moore, and Jones [41], we specifically name race and racism, rather than categorizing it with diversity.

Data Visualizations By Journal Fig. 2 shows the number of publications sorted by the seven journals we examined. JBTC significantly leads in publications, with TCQ and CDQ trailing not far behind. JBTC leads in publications in part because of its COVID-19 special issue [37]. Apart from the two aforementioned special issues, the reasoning behind this distribution is not immediately apparent. It is noteworthy that the two journals, whose focus revolves around business communication, bookend the dataset with the largest and smallest respective number of publications.

While the types of data visualizations discussed were somewhat evenly distributed across the

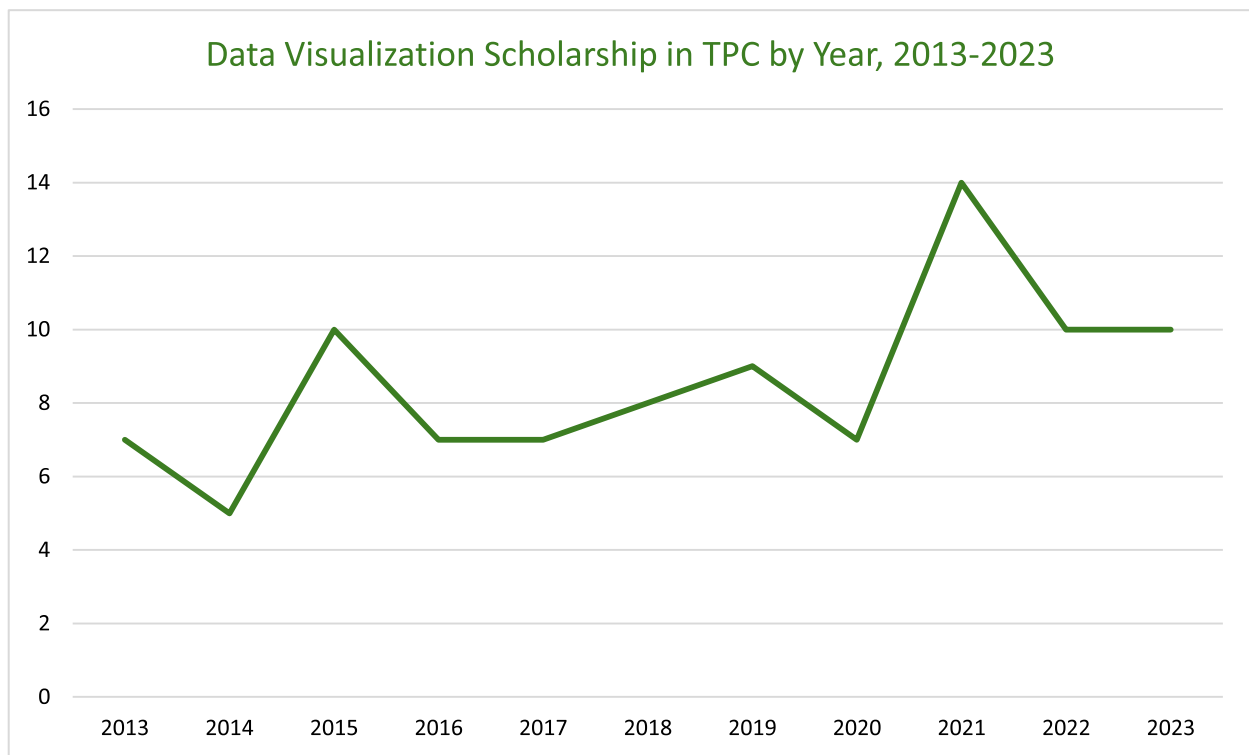


Fig. 1. Sorting data visualization scholarship by year shows an uptick in publications prompted by the COVID-19 pandemic.

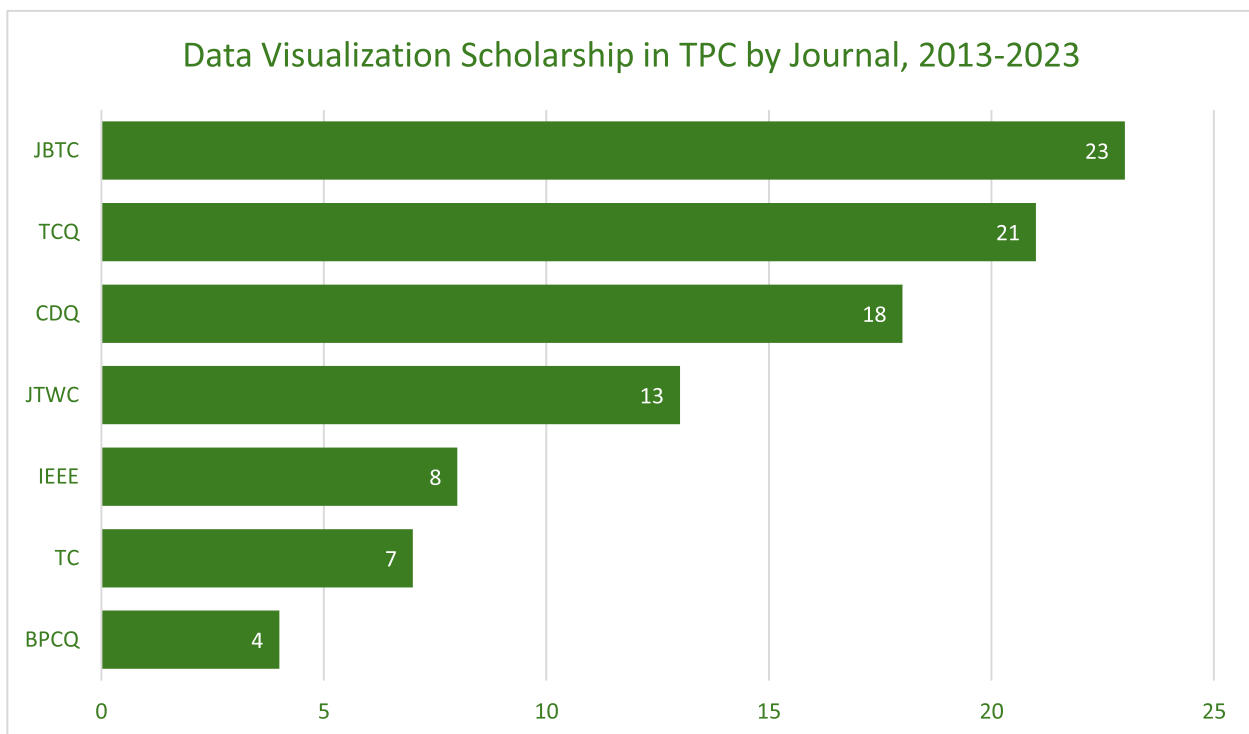


Fig. 2. When measured per journal, data visualization scholarship was published most in JBTC and least in BPCQ.

journals in our dataset, notable patterns emerged. *CDQ*, for example, published a high number of articles focused on mapping. Similarly, *JBTC* led in publishing work related to dashboards with five publications. These patterns suggest that while all journals engage with diverse forms of visualization, certain venues may serve as hubs for specific types. The next section explores these data visualization types in greater detail.

Data Visualization Types We were not surprised to see infographics, bar charts, and line graphs in the top five codes. Infographics were our most commonly found data visualization type. This may be in part because of the broad way that infographics are defined or the way that we defined them. For instance, we coded traditional infographics, such as W.E.B. Du Bois' infographics [42], alongside items, such as science illustrations [43] and the US Census [44]. Cartographic maps were the next most cited data visualization type, focusing on geospatial representations of phenomena, such as sea level rise [45] or flu spread [46]. Our third and fourth most cited data types were more traditional genres: line graphs [46] and bar charts [17]. Surprisingly, the fifth most common type was dashboards, which we defined as online sites with at least one data visualization that was updated in real time [47].

Coding Results: Engagement With Data

Visualizations In this section, we think more deeply about what our codes indicate about the state of the field. We interpret our findings by examining the frequency of each code and their occurrences across articles, using these patterns to identify and track which themes technical communicators have focused on in their work. We found it less important to employ statistical significance to look at correlations between codes but instead observed patterns between code pairings. In our discussion, we look at the top 15 (out of 28) most frequently used codes to see where the field has been and where the field can go.

The 15 most frequently used codes, found in Fig. 3, indicate the kind of subject matter, arguments, and contexts that the scholars in our dataset explore in their work. Leading our codes is “color,” which we noted when the rhetoric of color in data visualization became central to a scholar's analysis [15], [48]. “Risk” plays a large role in our dataset and comes up in a variety of different risk communication pieces, such as those that relate to sea-level rise and climate change [40] and public health [13]. In addition, codes, such as “social

justice,” indicate scholars' attention to the way that data visualizations either work with or against marginalized communities [15], [44], [49]. Other codes, such as “misinformation” [17], further described scholars' critical focus on how data visualizations have been used to manipulate readers.

Color The use of color [8], [15], [48], [50], [51] has been looked at in a variety of health, medical, or disease data visualizations [52], [53]. Because matters related to design are so central to scholarship in data visualization, this code likely appeared so frequently because of how often scholars discussed design strategies for the visualizations they were writing about. Since we drafted the codebook, we considered adding codes, such as “design” or “aesthetics,” but later decided that these codes could be used so often that they might lose their meaning. However, the “color” code indicates a specific component of design that scholars drew particular attention to in the last decade.

Risk In TPC, data visualizations were a common way to communicate risk to lay audiences, particularly through mapping. Disease maps, DIY risk assessment [46], or climate change maps [54] can effectively communicate risk of a variety of health and environmental hazards. As the threat of climate change has increased, data visualizations have become a popular way that people learn about potential risks affecting themselves and their local communities.

User Experience Work tagged with UX looked at the creation or design process of data visualizations and tends to give practitioner advice and guidelines for UX professionals. The UX code is closely related to the health, interactivity, and social justice codes because a UX lens facilitated effective health interfaces [52], engagement with interactive risk tools [55], and gave users information about social justice issues, like the climate crisis [45].

Interactivity Scholarship tagged with the interactivity code focused on how data visualizations supported user agency and information access through data visualizations that allow users to directly engage with and manipulate data displays [55]. Work on interactive data visualizations that gave users access to real-time data reflected a kind of transparency involved by letting users have access to unprocessed data [56]. Some scholars also explored how interactivity supported accessible design practices [57].

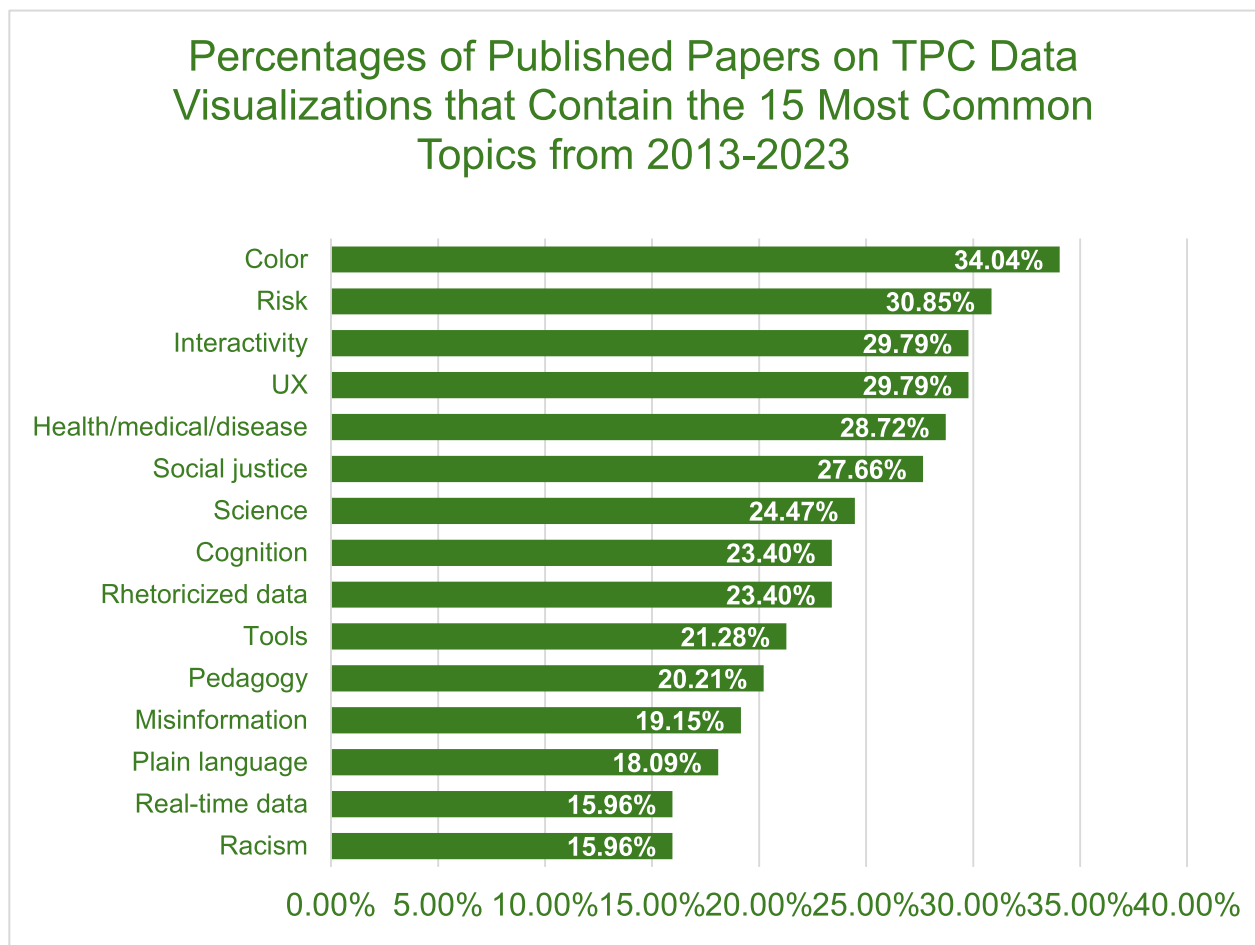


Fig. 3. Color, risk, and UX were the most frequently used codes in our dataset.

Currently, the extant literature suggested that interactive data visualizations offered ways for users to interact with data that afford them more transparency and choice than static representations.

Health/medical/disease Many scholars within TC were concerned with the way that data visualization related to health and disease. Whether scholars looked at health visualizations from a historical angle [51], a contemporary angle [58], or from a practitioner perspective [52], scholarship tagged with the health, medical, or disease code highlighted ways that data visualizations can either promote or prevent access to health information.

Social Justice The social justice code allowed us to obtain a broad view of how scholarship in the field has uncovered the ways that data visualizations have worked to empower or disempower marginalized communities [41]. Scholarship related to social justice has looked at

how data visualizations have constructed race and have enacted racism [59] and has explored the relationship between racial oppression and the environmental crisis [60] and health crises [61]. This work varied widely in the methodological approach, as some utilized archival methods [61], while others employed narrative and storytelling techniques [62], [63]. This work critically demonstrated where social justice inequities historically came from, how they manifested in the world (and in data visualizations more specifically), and how these patterns could be perpetuated or disrupted.

Science In TPC research, data visualizations often helped make scientific information more digestible for lay audiences. We noticed that our science code was frequently paired with some of the other aforementioned codes. For example, we used the science code with the health, medicine, or disease codes when the article explored the underlying scientific theory of these issues [53],

[58]. Our science code was also strongly tied to work regarding interactive climate visualizations [40], [64] and risk [65]. Science was also closely related to our “pedagogy” code for the ways that scholars wrote about data visualizations in STEM classrooms [66], [67], [68], [69].

Rhetorized Data The rhetorized data code highlighted scholarship that critically engaged with the way that data have been presented to emphasize a particular perspective, instead of data as purely neutral or objective. This code paired with pedagogy in pieces that focused on teaching ethical design practices within the classroom [7]. This code was also connected to the health, medical, or disease code in public-health pieces [20], especially with COVID-19 data visualizations [21], [22], [23]. This code was often paired with misinformation because of the way that data were utilized to emphasize a particular narrative to users [7].

Cognition We used the cognition code when scholars were interested in the way that people psychologically experienced data visualizations, as cognition connects with both misinformation and accessibility for neurodivergent users. We applied this code to think about how data visualizations could deceive or be used to spread misinformation [10], [19] or to think more broadly about how users processed data visualizations in terms of complexity, detail, and message [40].

Tools Our tools’ code was used when scholars centered existing technologies used in the creation of data visualizations [70], new technologies [71], or broadly covered the use of tools within the data visualization creation process [72]. We also observed the tools’ code to be closely related to the aforementioned cognition code when scholars employed technology, such as eye tracking [19]. We expected to see the tools code show up in tutorial work, but we noticed significantly fewer tutorials in our data set compared to participant and case studies.

Pedagogy Our dataset demonstrated that data visualizations could be powerful pedagogical tools. Whether scholars looked at data visualization from a social justice lens [73] or a rhetorized data lens [7], many scholars were interested in teaching to disrupt unethical design practices. We also noticed that pedagogy was linked with our “tools” and “UX” codes, suggesting that pieces focused on practical application.

Misinformation We tagged scholarship focused on how data visualizations might be complicit in the spreading of false information with the misinformation code. We noticed that this code was particularly prevalent on pieces that looked at issues related to health [17] and science [74]. While our “rhetorized data” code looked at how data were used to emphasize a particular narrative, the misinformation code looked at visualizations containing false or misleading claims regardless of intent.

Plain Language Plain language sought to make technical writing more accessible for broad audiences. When applied to data visualization contexts, plain language further facilitated audience understanding in work centered on science [75], risk [65], and health [76]. We also found plain language closely related to pedagogy and business communication [77] because of its relevance to business genres.

Racism The racism code explored scholarship about how data visualizations have enacted and perpetuated racism in historical [50] and contemporary [15] contexts. These sources were often linked with the social justice code, but where that code explored systemic oppression broadly, this code specifically looked at racism as a form of systemic oppression. We answered the call by Jones, Moore, and Walton [41] to specifically name racism and racial oppression, rather than combining it with social justice and/or diversity.

Real-Time Data Articles coded with “real-time data” explored data visualizations that presented users with newly collected information with little processing or delay. Real-time data gave transparency to data-collection and interpretation processes in risk [45] and health [46] contexts to expand user agency [78]. Many, but not all, of these articles were linked to social justice because they explored expanding access to information for users.

DISCUSSION

In this section, we take inventory of the patterns that emerged from our analysis and think about how data visualization practices and priorities have shifted over time. We highlight unexpected answers to the research questions, discuss notable trends in the uptake of specific visualization types, such as cartographic mapping, and examine how visualizations have been used to communicate risk and uncertainty. Finally, we explore how data visualizations can be rhetorically positioned in

contexts related to health, risk, accessibility, and social justice.

Taking Inventory When answering these research questions, we observed that codes, such as social justice, risk, health, medical, disease, and misinformation, were consistently tagged throughout the decade. These codes indicate that much of the work performed on data visualization has comprised human-centered work focusing on safety, well-being, and ethics. Scholars use their expertise to respond to crises, such as the COVID-19 pandemic, by advocating for better information practices, combating misinformation, and influencing health policy in real time. Scholarship on data visualizations was especially present in special issues of *JBTC* and *TCQ*. Surprisingly, *BPCQ* had only two examples. Considering how important dashboards are for communicating climate change and health decision making, we anticipate more scholarship on these topics. Misinformation was also very connected with the COVID-19 pandemic in 2020 and 2021; however, the spread of misinformation and disinformation has only accelerated since. More empirical work that relates perception, cognition, and literacy [10], [19] to accessibility would only serve to strengthen research on misinformation.

Changing Codes Over Time We believe it is important to contextualize coding shifts in relation to key scholarly and political events of the past decade. However, making direct comparisons across time presents challenges, as our dataset spans 2013–2023, resulting in uneven periods for analysis. For instance, when examining the impact of the social justice turn in 2016 or the COVID-19 pandemic in 2020, we can only compare postevent scholarship to a much smaller pre-event period, making it difficult to determine whether observed increases or decreases in codes reflect trends or simply differences in the length of each period. Despite this, we note relevant changes we observed in the last decade.

While data visualizations related to health, medicine, and disease had been a consistent area of inquiry in our dataset before the 2020 COVID-19 pandemic, we observed a substantial uptick of this code from 2020 to 2023, with publications, like the special issue from *JBTC* [37]. In addition, we noticed a gradual increase in work tagged with the social justice code after 2016 (two articles before 2016 and six articles between 2016 and 2019), and an increase after 2020 (13 articles from 2020 to 2023). Relatedly, we noticed a gradual increase in

work tagged with the racism code after the social justice turn (no articles before 2016 and five articles from 2016 to 2020), and another gradual increase after 2020 (four articles from 2021 to 2023). These patterns suggest that major social and political events have influenced the field's growing attention to issues related to health, justice, and race.

Cartographic Mapping as Data Visualization Type

Our dataset reveals that cartographic mapping remains a prominent and evolving data visualization strategy, particularly for representing complex societal issues related to health, risk, and the environment. Emerging from the grand tradition of Booth's poverty maps in London [79], we noticed that mapping is still a critical way to illustrate societal issues related to health, disease, risk, and the environment. To put it simply, when scholars mapped data, they generally mapped the effects of climate change, issues of geographic racism, or the locations of specific diseases or epidemics.

Articles in our dataset illustrate how maps operate not only as neutral tools but also as rhetorical documents that reflect and shape public discourse. For example, we observe mapping as interrogated from institutional sources, such as the 2011 Texas District Map [15], or observed from grassroots movements, such as user-contributed flu maps [46]. These examples demonstrate that cartographic maps can be used as both documents of authority and as sites for advocacy. Maps are a sense-making tool for wicked problems, especially those related to risk.

Visualizing Risk Communication Risk visualization has emerged as a subfield within TPC data visualizations [80]. The articles in our dataset that visualize risk typically rely on mapping to communicate data about health and disease, or the environment. Maps can lend insight into what it means to put people and land at risk of harm [60]. TPC scholars have been careful to think through issues of participatory design, asking which people and communities are centered when creating visualizations [62], [80], [40] as vectors for storytelling.

Scholarship on maps and visualizations also helps us better understand embodied risk, whether through storytelling [14] or geospatial information [13]. In the latter example, Wellhausen [13] shows how risk maps shifted to vaccine maps, demonstrating how the focus on embodied risk

becomes the call to action. Maps present a multivalent and nuanced sense of risk for users through localization: the immediacy of understanding one's state, municipality, or neighborhood in relation to an issue [8]. Communicating risk is a wicked problem that TPC data visualizations are uniquely positioned with which to grapple, as we work to increase visual risk literacy [25].

Rhetorized Data and What We (Cannot)

Assume to be Objective Articles coded with "rhetorized data" often examined genres traditionally perceived as neutral or objective, such as maps and demographic data representations. Key themes emerged around how these visualizations, including maps depicting racial gerrymandering [15] or historical visualizations of Chinese immigrant populations [59], inherently involve rhetorical decisions that shape the presentation and interpretation of data. In addition, analyses of demographic collecting documents, such as census data [44] and COVID-19 demographic visualizations [21], emphasized how seemingly objective data presentations influence and impact real people's lives through implicit or explicit positioning. These findings highlight the necessity of critically examining the rhetorical dimensions of data in data visualization genres that ostensibly present themselves as neutral.

Accessibility is the Future We were surprised that only four articles in our dataset discussed the importance of accessibility in data visualizations, which was not always connected to the social justice code. Strantz [4] called on technical communicators to integrate web standards for accessibility when visualizing data. Winn [81] detailed accessible way-finding practices; Heilig [82] connects campus maps with interactive accessibility. Conway, Oppegaard, and Hayes [57] detailed their exciting work to create audio descriptions of maps for blind people. To meet disabled users' needs, research on accessibility must continue to share knowledge about interactivity, gestural accessibility for smartphones and tablets, improved readability and attention to cognition, and audio or alternative text descriptions.

CONCLUSION

From 2013 to 2023, data visualization scholarship addressed critical issues related to ethics, health and medicine, and risk communication, drawing on both scholarship from TPC and interdisciplinary

work. We would like to highlight the need for more scholarship that expands current practices in accessibility and explicitly connects that work to social justice. We only found four works that explicitly foregrounded accessibility [4], [57], [81], [82], suggesting a significant need in the field, considering the reliance on visualizations in public-facing contexts. Accessibility might also be more fully integrated into work that discusses tools and pedagogy. Future research could explore not only how to make visualizations accessible, but also how accessibility concerns shape the development of visualization tools, UX, and methods for distribution.

Pedagogy in an Age of AI and Misinformation

As we face unprecedented ethical challenges with artificial intelligence, TPC faculty must teach students about socially just design practices.

Data visualization ... does not just involve representing a given set of numbers; it involves selecting and rethinking the numbers on which the visualization is based. [7, p. 347]

Scholarship on data visualization can be a powerful way to teach audiences about complicated topics with direct implications for their health and safety. However, work across TPC underscores how easily visualizations can mislead users through misinformation, unethical design strategies, or the rhetorical framing of cherry-picked or incomplete data [17], [20], [21]. Since we continue to face the cumulative effects of climate change, accurate and ethical risk communication will become even more important to our field and our humanity. Future technical communicators will need to make skilled value judgments and rhetorical decisions about the data they work with to ensure that visual data not only supply facts but also empower audiences to make well-informed decisions about their health and safety. As a field, TPC can produce more pedagogical research on data visualization practices to prepare both faculty and future practitioners to face these challenges.

Limitations and Future Directions Any study of this size and scope has its limitations. With a dataset of 94 academic articles ranging from five to sometimes more than 20 pages, finding a balance between the number and type of codes used to describe the complexity and nuance of each published work is an inherent challenge. We faced this challenge by coding individually and cross-checking for consensus. Despite our best efforts, certain nuances may not have fully been

captured. The scope of this study also made it difficult to keep up with new scholarship that was being published alongside the research and writing for this project.

Overall, we captured what we could while being transparent for replicability and applicability. We are interested in work that continues to grapple with the enormous but undoubtedly useful task of creating a taxonomy of data visualization scholarship to better understand what is and is not being given scholarly attention. Expanding and refining such categorization could provide valuable insights into the evolution of the field, highlight emerging areas of scholarly interest, and identify respective opportunities in the literature.

Final Thoughts Reflecting critically on the topics, methodologies, and voices prioritized in TPC data visualizations gives us a chance to see how our work might better align with our scholarly values and what we can uniquely add to interdisciplinary work. TPC data visualizations frequently appear in contexts related to ethics, health, and risk, demonstrating a strong commitment to tackling societal issues. However, our findings revealed an underrepresentation of work dedicated explicitly to issues surrounding accessibility and cognition. We are excited that the field values social justice within the contexts of science, health, medicine, and climate change and hope that scholars turn their attention toward issues related to accessibility.

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